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PROJECT THESIS

Toxicity Prediction of Plastic Chemicals

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Contents

1 Introduction **2**

2 Methods **4**

2.1 Dataset 4

2.2 Model development 6

2.3 Computing environment 8

3 Results **9**

3.1 Grid search 9

3.2 Data processing 9

3.3 Performance 11

4 Discussion **14**

4.1 Dataset limitations 14

4.2 Model performance metrics 15

4.3 Generalization limitations 15

4.4 Outlook 16

References **17**

S1 Code availability **21**

S2 Additional Figure **21**

Declaration of Academic Integrity **25**

1 Introduction

The impact and scale of plastic production and consumption on the environment is steadily increasing. In 2020, the world-wide cumulative primary plastic production surpassed ten billion tonnes. Only around 10% of global plastic waste has ever been recycled. Of the remaining 90%, 14% have been incinerated, producing greenhouse gas emissions and releasing gaseous pollutants in the process. The remaining 76% stay intact, leaching toxic chemicals and microplastics into the environment (Walker 2025).

A multitude of polymer groups are used in industrial settings, which can be differentiated based on their polymer backbone. The largest groups in commodity plastics are polyethylene (PE; 36%), polypropylene (PP; 21%) and polyvinylchloride (PVC; 12%), followed by polyethylene terephthalate (PET), polyurethane (PUR) and polystyrene (PS) (less than 10% each). In total, these groups add up to 92% of all plastics ever produced (Geyer et al. 2017).

The wide variety and usability of plastic products stems from the immense range of chemicals that are added during the manufacturing process for different reasons. Without additional chemicals, polymers are not broadly usable or durable (Wagner et al. 2025). Additives aid in manipulating the plastic products properties. Plasticizers, for example, are widely used to make the endproduct more flexible, or colorants, which give the endproduct its color (Hahladakis et al. 2018). Processing aids, such as catalysts and solvents, help during the production and processing of polymers. Plastic chemicals can also be unintentionally added. Such non-intentionally added substances (NIAS) can be categorized into side products, breakdown products and contaminants. The extent to which manufacturers are aware of the presence of these substances in the end product varies. It is evident that the knowledge and classification of NIAS is, in general, limited, due to a lack of both information and toxicity testing. This is partly due to complex structures such as oligomeres being present (Geueke 2018).

Plastic chemicals are used to give the end product specialist functions, while often being the cause of the end product having toxic properties. Humans are at risk of plastic chemical exposure during every step of the products life cycle either directly (e.g., plastic straws, plastic childrens toys) or indirectly (e.g., ingestion of food packaged in plastic, inhalation of plastic chemicals from air). Plastic chemicals have been linked to many adverse health effects, such as nonylphenol (a plasticizer) and bisphenol A (used as a starting material for polymer synthesis) being endocrine disruptors (Pérez-Albaladejo et al. 2020). Plastic additives bonds to polymers are weak and therefore pose risks of them easily being released into the environment at any stage during the plastic life cycle.

The reusability and recycling capacity of the plastic product is influenced by the chemicals as well, as high quality recycling products can often only be produced from well-separated clean recyclates, that is, recyclates free of most plastic chemicals. Additionally, chlorinated plastic chemicals can lead to corrosion on recycling machinery. Despite all of these problems, plastic chemicals remain widely unregulated and undocumented and often lack proper hazard information (Wagner et al. 2025).

PUR is a highly diverse group of synthetic thermoplastic polymers created via polycon-

densation reactions between isocyanates and compounds containing hydroxyl. The wide range of possible reactants forming polyurethanes leads to their usage in a multitude of industries as sealants, coatings, elastomers, and foams. Recycling them is of importance considering the ongoing depletion of fossil fuel reserves (Rossignolo et al. 2024).

Plastic products are primarily recycled mechanically or chemically by depolymerization. Mechanical recycling focuses on thermoplastics such as polyethylene (PE) and polyethylene terephthalate (PET), which consequently are the most recycled plastics.

The following report predicts the toxicity of plastic chemicals present in the dataset released by Wagner et al. 2025 by training a machine learning model using XGBoost (Chen and Guestrin 2016). They present 16 325 known plastic chemicals and their properties compiled from multiple open databases. Only around 6% of them are internationally regulated, and more than 10 000 are missing any hazard information at all. This motivates a structure-based toxicity prediction approach. A significant implementation challenge is posed by the significant class imbalance of the available hazard information, which is addressed via a combination of resampling techniques and a custom loss objective.

2 Methods

2.1 Dataset

The PlastChem database (Wagner et al. 2025) is the largest publicly available plastic chemical database to date. It was built to address the scarcity and scattering of chemical data on plastic chemicals, a problem faced by policymakers and researchers in the plastic sector. 17 932 plastic chemical entries that are present in plastic products were compiled and identified. The database was created through a systematic collection and harmonization of information from scientific and regulatory sources. In addition to identifiers, it contains information about chemical properties, regulation, hazard information, structural groups, production and functions across multiple plastic subgroups. The authors identified chemicals of concern based on their hazard information, which was determined on four criteria: persistence, bioaccumulation, mobility and toxicity. These four criteria were transferred to a single hazard score. For more information on the hazard scores, see Supplementary information of Monclús et al. 2025 and Figure S1.

Across the entire database, around 25% of all listed chemicals have been assigned a hazard score. The ten largest polymer groups, that is, the ten groups with the most chemicals assigned to them, each contain at least 400 chemicals of concern. This work uses the PlastChem database (version 1.1, Wagner et al. 2025) as the source for both molecular structures and hazard labels for training a classification model.

For the purpose of this project, only compounds meeting both of the following criteria were used for model training and evaluation:

1. a canonical SMILES³ string was present, allowing computation of Morgan molecular fingerprints⁴ (Rogers and Hahn 2010) as a model input, and
2. the hazard score label was present, allowing its direct usage as a toxicity label.

Entries failing either criterion were excluded, as neither a valid molecular representation nor a usable target label could be constructed for them. Compounds for which fingerprint generation failed (e.g. due to invalid SMILES) were removed as well. This resulted in 4 748 compounds that were deemed usable for the model training. Finally, hazard scores were binarized into hazardous and not hazardous, by merging the intermediate and high-hazard categories into a single class.

Morgan fingerprints of length 2048 were computed for all remaining samples. Additionally, the molecular weight, topological polar surface area⁵ and complexity⁶ were appended onto the fingerprints as complementary information. These three features represent global molecular properties that are not encoded in Morgan fingerprints. Per-sample dimensionality was reduced using a principal component analysis (PCA), a linear dimensionality reduction technique that identifies the n components in the data that are linearly uncor-

³Simplified Molecular Input Line Entry System; a representation of molecular structures as ASCII-strings

⁴binary encoding of molecular structures

⁵sum of surface contributions of all polar atoms

⁶algorithmic complexity score calculated by PubChem

related.

The resulting dataset is strongly imbalanced toward the hazardous class across all polymer types. In total, the hazardous class represents 97% of the data. Figure 1 shows the imbalance per polymer type. Across all ten most abundant polymers, the hazardous class makes up between 95% and 98% of compounds, with PET and PP being associated with the largest absolute number of chemicals of concern and PC (polycarbonate) and ABS (acrylonitrile butadiene styrene) with the lowest amount. As this data is subject to sampling bias and only represents the known plastic chemicals, one should not think of polymer classes with more toxic chemicals associated to them as more toxic than others.

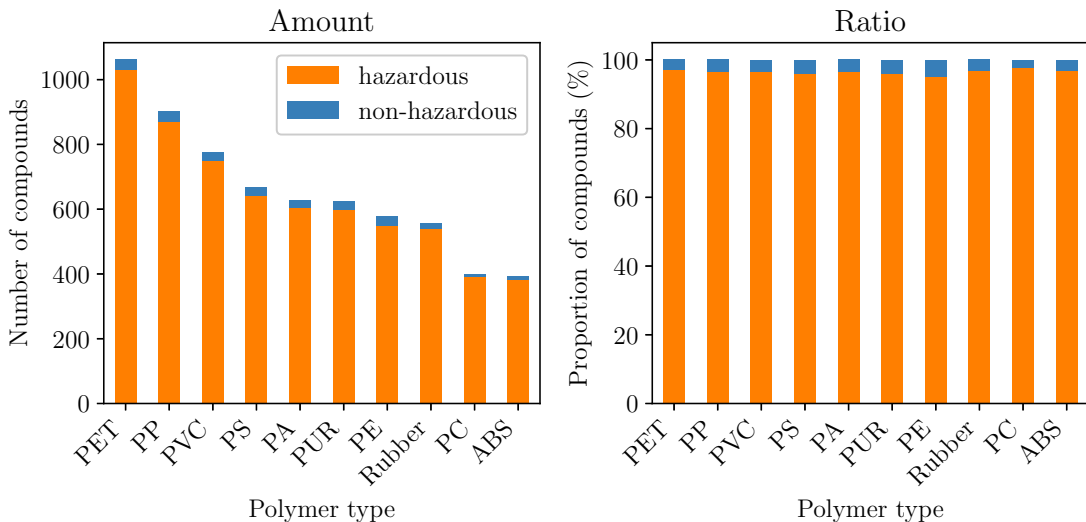


Figure 1: Amount and ratio of hazardous and non-hazardous compounds in the dataset displayed by polymer type for the ten most abundant polymer types present.

This pronounced imbalance is the primary reason resampling was incorporated into the training pipeline described in section 2.2, and why balanced accuracy rather than raw accuracy was used as the model selection criterion during hyperparameter search.

Another complexity the dataset presents is its high dimensionality. This is invariably the case when dealing with chemical data represented as high-dimensional sparse fingerprints. Figure 2 visualizes the complexity of the data using a two-component PCA projection of the Morgan fingerprints, in which the two components capture just 8.56% of total variance.

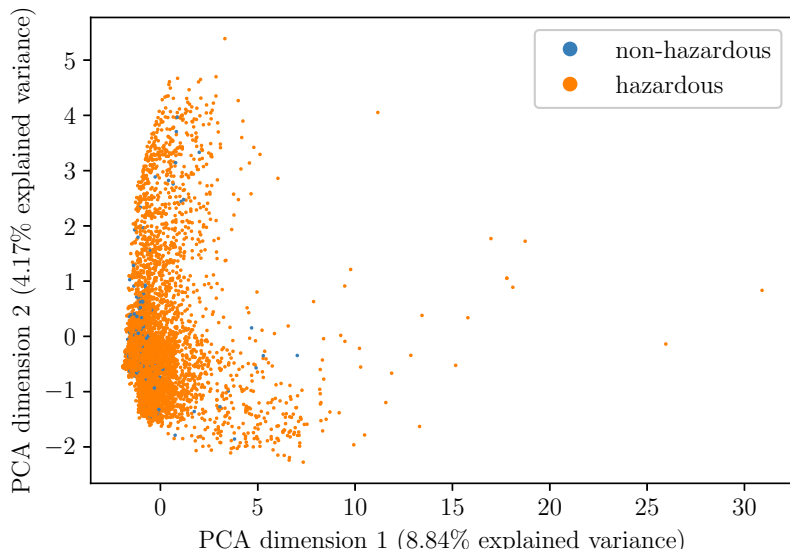


Figure 2: Two-dimensional principal component analysis of dataset represented as molecular fingerprints, explaining 8.56% of total variance.

Hazardous compounds (orange) are distributed broadly across the whole projected space, while non-hazardous compounds (blue) are comparatively rare, with no clear linear separation visible in the first two principal components. Due to this lack of linear separability in a low-dimensional projection, we decided to use the XGBoost classification algorithm as it was shown in the literature to handle high-dimensional data well.

2.2 Model development

The aim of this project is to build a binary classifier, which, given the molecular structure of a plastic associated chemical encoded as a fingerprint, predicts whether the compound is hazardous or not hazardous. As described in section 2.1, the dataset is highly imbalanced and not easily separable. In order to build a reliable classifier, the decision was made to binarize the hazard labels, as multi-class classifiers are even more susceptible to imbalance. The classes were binarized by joining all labels with various degree of hazardousness into one, while the non-hazardous class was left as is.

The dataset (the remaining 4 748 compounds with a valid SMILES-derived fingerprint and a hazard-label, see section 2.1) was split into two disjoint sets, with 70% of the data used as the exploration set for a 5-fold nested cross-validated hyperparameter search, and the remaining 30% as a test set. All results in section 3 are reported on the independent test set. The splits were stratified on the target hazard label in order to preserve the previously described class imbalance, and a fixed random seed was used throughout for reproducibility, as implemented in the scikit-learn package (Pedregosa et al. 2011).

To prevent data leakage, preprocessing and classifier training were combined into a single pipeline, ensuring that all steps are re-fit independent on each training fold rather than the

full dataset. A nested cross-validation approach was taken to jointly determine the best parameter values for the classifier and preprocessing steps and to obtain an unbiased estimate of the resulting model selection procedure’s generalization performance (Cawley and Talbot 2010). The inner loop performs an exhaustive grid search that fits and evaluates the pipeline via five-fold cross-validation for each combination of hyperparameters. The combination achieving the highest mean balanced accuracy across the folds is selected. This inner search is repeated independently of each other across five outer folds, whose held-out balanced accuracy scores are not used during hyperparameter selection but serve as the unbiased performance estimate reported in Section 3.3. The final hyperparameter configuration reported in Table 1 is obtained separately, by running the same inner grid search again on the entire exploration set, independent of the outer nested CV-loop.

First, a Principal Component Analysis (PCA) was applied to transfer the 2048-bit large binary fingerprints into a continuous lower-dimensional representation. Then, the pipeline applies the Synthetic Minority Over-sampling TEchnique algorithm (SMOTE, Chawla et al. 2002) combined with Edited Nearest Neighbors (ENN, Wilson 1972) undersampling. This approach oversamples the minority class by interpolating between existing samples in order to balance the class ratio. Subsequently, the sample space was cleaned using ENN, a technique that removes samples if any of its nearest neighbours are not from the same class. The final classifier is an XGBoost (Chen and Guestrin 2016) classifier employing a histogram-based tree method.

Morgan fingerprints are high-dimensional, sparse binary vectors, in which distance-based methods such as ENN and SMOTE are known to perform poorly due to the curse of dimensionality (Altman and Krzywinski 2018) especially considering the dataset size. Therefore, PCA was applied first to obtain a lower-dimensional, dense representation in which relationships between neighboring molecules are more meaningful. The optimal number of components for the PCA was determined during the grid search.

SMOTE was selected instead of simple random oversampling or class weighting because it generates synthetic minority-class samples by interpolation rather than duplication, which reduces the risk of overfitting the model to repeated examples. As a way to counteract the synthetic points in ambiguous or overlapping regions of the feature space introduced by SMOTE, it was followed by an ENN step to remove noisy majority-class samples near the decision boundary. Combining SMOTE and ENN follows the recommendation for a combined resampling by Batista et al. 2005. The combined scikit-learn implementation SMOTEENN (Lemaître et al. 2016) has a parameter to set the preferred ratio of minority class over majority class samples after resampling, which was optimized in the grid search.

XGBoost was used as the classifier due to its strong benchmark performance on tabular and molecular fingerprint data (see Ding et al. 2021 for an example). It employs a gradient boosting approach in which decision trees are added iteratively to improve the ensemble’s predictions on the training data. The final prediction for each sample is determined based on the combined predictions of all decision trees in the ensemble. The tree method "hist" was chosen due to its computational efficiency. Further, the maximum delta step parameter is set to 3 in order to restrict the weight of each tree, another measure taken to prevent overfitting on the majority class.

Additionally, a custom focal loss objective implemented by C. Wang et al. 2020 was

adapted for the scikit-learn API and used in the classifier. Focal loss was proposed with the goal to address binary class imbalance by reducing the loss allocated to well-classified samples (Lin et al. 2017). The focusing parameter was set to 2, as the authors described this value to work best. Because of time and implementation constraints, this parameter was not optimized in the grid search.

Finally, the four following parameters were optimized during the grid search:

- the maximum depth of each tree, restricting the size of each decision tree
- the learning rate of the classifier, restricting the impact of each new tree on the prediction
- the number of estimators, that is, decision trees forming the ensemble, and
- the subsample ratio of columns for each level, a further measure to prevent overfitting by reducing the number of features seen by the classifier during each tree level.

The results for each parameter that was optimized in the aforementioned grid search are displayed in section 3.

2.3 Computing environment

The model training was conducted on a compute node of the Friedrich-Schiller-University high performance computing cluster ‘Draco’. The node is equipped with two Intel Xeon Gold 6326 CPUs with 2.90 GHz. The system has a total of 32 physical CPU cores or 64 logical CPUs. For the training, 64 CPU cores were allocated with 32 GiB of memory. All model training was executed using Python 3.14.6. The primary software libraries used for data processing and machine learning included NumPy 2.5.0, pandas 3.0.3, scikit-learn 1.9.0, XGBoost 3.3.0 and imbalanced-learn 0.14.2.

3 Results

3.1 Grid search

An nested iterative grid search was performed, progressively narrowing the search range around promising values chosen from initial exploration runs (Table 1). The final configuration selected the following values:

- n_components=160,
- sampling_strategy=0.3,
- max_depth=7,
- learning_rate=0.01,
- n_estimators=300, and
- colsample_bylevel=1.0,

achieving a cross-validated balanced accuracy of 0.616 (see section 3.3 and Figure 5) over five folds. Notably, n_compounds, sampling_strategy, max_depth and n_estimators were selected at the boundary of their respective search ranges. The search grid for the sampling_strategy parameter was not extended further, in order to avoid the introduction of additional noise.

Pipeline step	Parameter	Explanation	Grid values
PCA	n_components	No. of principal components	<u>160</u> , 190, 220
SMOTEENN	sampling_strategy	desired class ratio	0.1, 0.2, <u>0.3</u>
XGBoost	max_depth	max. depth of trees	3, 5, <u>7</u>
XGBoost	learning_rate	impact of new trees	0.001, <u>0.01</u> , 0.1
XGBoost	n_estimators	No. of decision trees	100, 200, <u>300</u>
XGBoost	colsample_bylevel	feature reduction	0.5, 0.75, <u>1.0</u>

Table 1: Results of iterative grid search, with chosen parameters underlined.

3.2 Data processing

As described in Table 1, the grid search for the optimal number of principal components resulted in 160. This number of components explained a total variance of 59.21%.

As shown in Figure 3, the cumulative explained variance ratio increases steeply for small numbers of components before slowing at around 500 and finally flattening at around 1250 components. This reflects the diminishing marginal information content of additional principal components once the components representing the majority of variance have been captured.

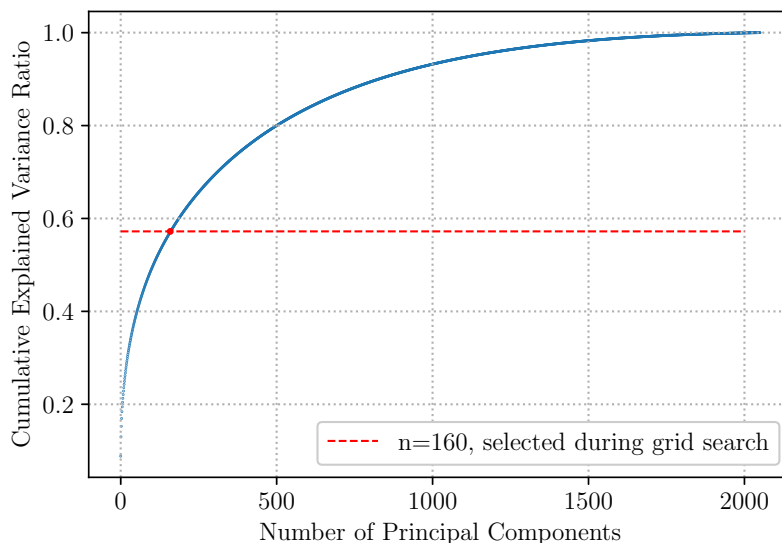


Figure 3: Cumulative explained variance ratio by principal components. Red marker indicates number of components selected by grid search (n=160, corresponding to 59.21% explained variance.)

Additionally, the influence of the appended molecular descriptors was investigated using the absolute factor loadings of the PCA compounds, displayed in Figure S2

Figure 4 illustrates the effect of the applied resampling strategy on the previously described class imbalance. Before resampling, the training set consisted of 3 323 total samples, with 109 labelled as non-hazardous, while the remaining 3 214 were assigned to the hazardous group. After resampling, the training set consisted of a total of 3 555 samples, 922 of which are labelled as non-hazardous, and 2 633 as hazardous. The imbalance between the classes is considerably reduced by the described approach. The resulting resampled dataset was subsequently used for the model training. The effect of the resampling on the model performance can be observed in Figure 5.

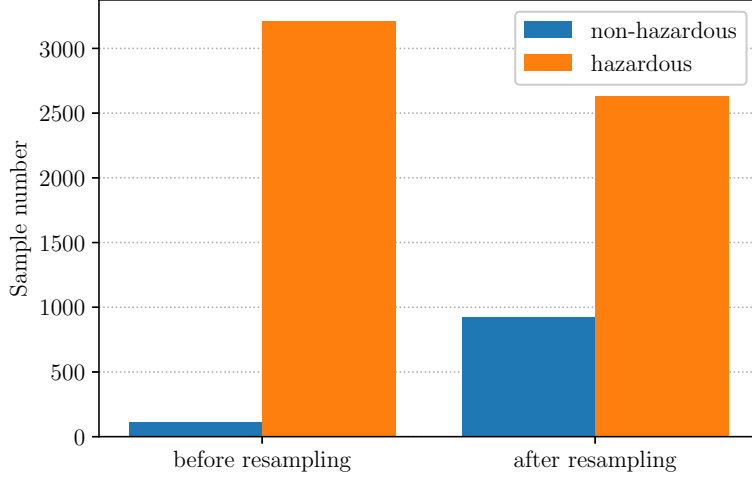


Figure 4: Effect of SMOTEENN resampling on the class distribution of the training set.

3.3 Performance

Figure 5 compares nested cross-validation performance between the pipeline with and without resampling (SMOTEENN). In both cases, the pipelines were trained on the full grid. Without resampling, balanced accuracy remains close to the 0.5 chance level across all five outer folds, suggesting this classifier mainly predicting the majority class. This is further supported by Figure S5. With resampling, the mean balanced accuracy scores increased, indicating it improves the classifiers ability to discriminate between classes. Notably, the performance varied between folds, ranging from around 0.454 to around 0.666. Fold 4 performs slightly worse with resampling than without. This reflects the increased sensitivity of the pipeline with resampling to which specific minority-class samples fall into each fold. The mean balanced accuracy as indicated by the black line reached 0.615. On the independent test set, a balanced accuracy of 0.616 was achieved. Overall, the results show substantial variation between the individual cross-validation folds, while the test set achieved a similar balanced accuracy score to the mean cross-validated score.

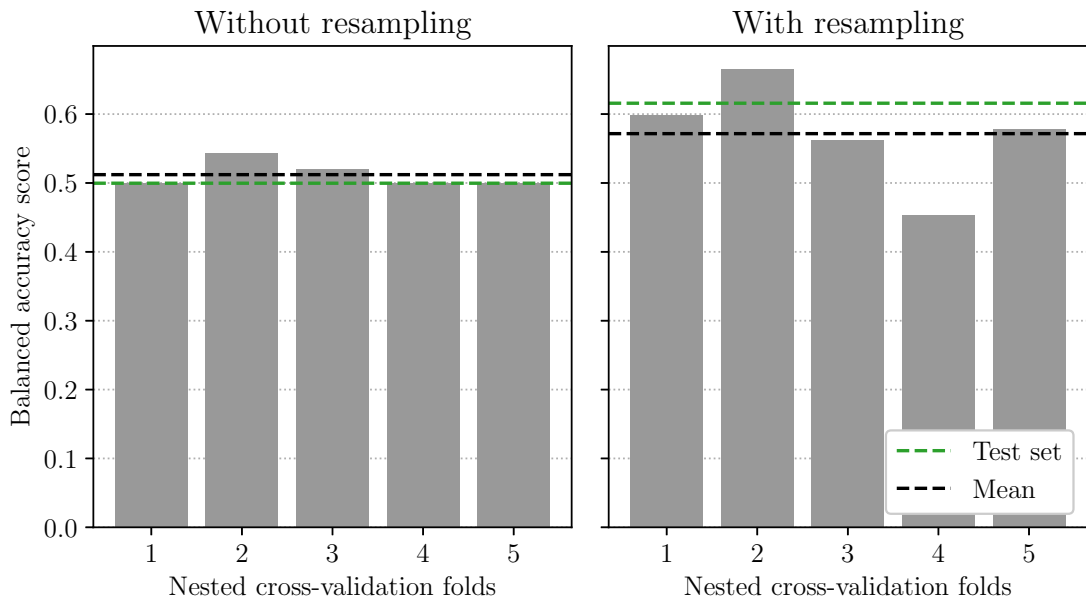


Figure 5: Nested cross-validation performance (balanced accuracy across five outer folds) of the pipeline trained with and without SMOTEENN resampling. Dashed black line indicates mean balanced accuracy across all folds. Dashed green line indicates balanced accuracy of test set.

The corresponding inner cross-validation scores, systematically higher due to their role in hyperparameter selection, are shown in Figure S3.

The model reached a balanced accuracy score of 0.616 and a ROC-AUC score of 0.723 on the test set. As seen in Table 2, the model achieved vastly different performances on the two classes. For the majority class, it performs well, achieving a precision of 0.97 with a recall of 0.96 and a F1 of 0.96. On the minority class, it severely underperforms, achieving a precision of 0.17, a recall of 0.28 and a F1 score of 0.21. The support for the minority class is 47, indicating that every classified compound has an effect of around 2% on the reported recall. The model appears to be strongly optimized towards the majority class. While the results presented below indicate its reliable detection of hazardous compounds, the performance for the non-hazardous class is notably low and appears to be a significant limitation of the model, further suggesting that minority-class predictions should be interpreted with caution. These measures are additionally supported by Figure S4.

Class	Precision	Recall	F1-Score	Support
non-hazardous	0.17	0.28	0.21	47
hazardous	0.97	0.96	0.96	1378

Table 2: Classification performance on the test set.

Figure 6 shows precision-recall curves for both classes on the independent test set using the final model. The hazardous class achieves an average precision (AP) of 0.98, while the minority class reaches an AP of 0.02, below its class occurrence of approximately 0.034. Section 4 discusses the difference between the two curves and how it impacts the model

reliability.

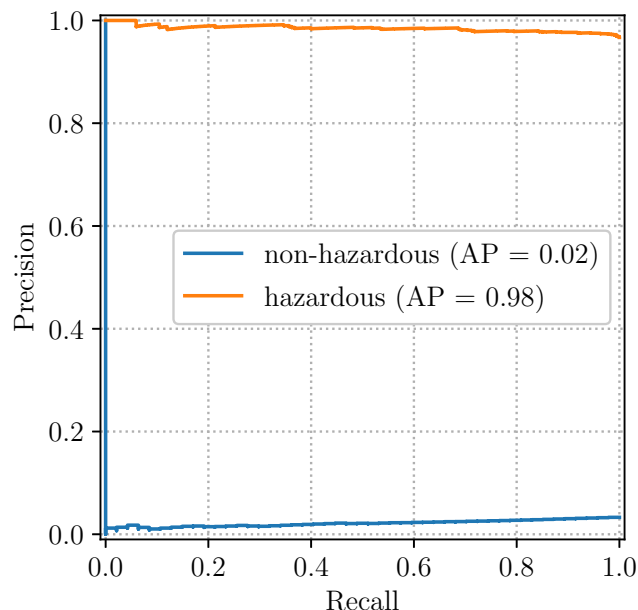


Figure 6: Precision-recall curves for both classes on the independent test set (n=1425, of which 47 are labelled as non-hazardous).

A comparison of the final model against a most-frequent dummy classifier (always predicting the majority class) can be seen in Figure 7. Notably, the model outperforms the dummy classifier in this measure, indicating predictive performance above chance. The discrepancy between the metrics displayed in Figure 6 and Figure 7 is further examined in section 4.

See Figure S6 and Figure S4 for confusion matrices of a dummy classifier and the trained model’s performance on the test set.

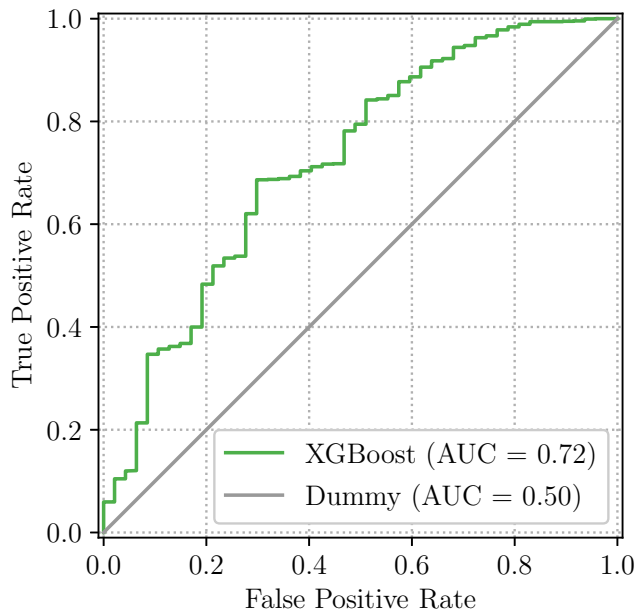


Figure 7: ROC curve for the final XGBoost model on the independent test set, compared against a dummy baseline.

4 Discussion

4.1 Dataset limitations

The main limitation of this project thesis is the limited data availability. The previously described imbalance of the class distribution leads to overfitting to the majority class, which the results in section 3.3 illustrate. The model is unable to learn the structure of non-hazardous compounds given their limited availability in the training data. Resampling strategies such as SMOTEENN improve the models performance but have to be viewed critically, as it cannot be verified whether synthetic interpolations result in naturally occurring (unknown) compounds. Representing the compounds using morgan fingerprints is limited to the fingerprint size. Different fingerprinting techniques have different advantages and may be more useful for harnessing the toxicity signal.

More data availability would increase the predictive power, but the plastic industry has no incentive to release data that could potentially harm their sales. The scarcity present in the dataset is a property of how the underlying data was generated. As discussed in section 2.1, more than 10 000 of the 16 325 plastic chemicals present in the PlastChem database lack any hazard information at all. Additionally, there is a bias in the data that is already released: the more toxic a compound is, the more likely its toxicity will be investigated. As a result, the present labels reflect regulatory and industrial testing priorities instead of the true frequency of hazard present in plastic chemicals. This is related to a broader issue already raised in the Introduction: plastic product manufacturers are not required to disclose toxicological information for plastic chemicals (Geueke 2018), so the class imbalance observed is symptomatic of the regulatory shortcomings rather than

a sampling issue.

4.2 Model performance metrics

The data imbalance also has a significant impact on the reliability of the model performance metrics. The per-class performance is asymmetric: the model correctly identifies 90% of hazardous compounds, while simultaneously only 12% of the compounds it predicts as non-hazardous are correctly identified. As the minority class consists of only 24 samples, every sample shifts the recall by around 4%. The performance metrics for the minority class should therefore be interpreted as noisy estimates rather than robust measures. This is also underlined by the cross validation performance across the five folds displayed in Figure 5.

Additionally, Figure 6 and Figure 7 differ in their conveyed message due to the imbalance. The precision is reported at a threshold of 0.5 while the average precision (AP) reported in Figure 6 is averaged over the precision at all classification thresholds, weighted by the recall gained at each. For the non-hazardous class, the precision-recall curve stays close to the class occurrence over most of the range. As a result, the average precision (AP) is only 0.02, which is considerably lower than the reported precision of 0.12 obtained at a default threshold of 0.5. The threshold at 0.5 shows one rather favorable point compared to the otherwise weak curve. For the minority class, the model shows weak precision across thresholds. While the performance seems overly optimistic in the ROC-AUC curve, it falls short in the precision recall metric for the minority class. Partially, the reason for this divergence lies in the fact that the model was trained on SMOTEENN-rebalanced data but evaluated at a threshold not tuned for the imbalanced test set distribution.

Threshold rebalancing was considered but was omitted from this work due to time constraints. Considering the scale of the precision gap, its effect may also be rather small.

4.3 Generalization limitations

As novel plastic chemical compounds are synthesized linearly by adding active groups to already existing compounds, the chemical space of known compounds can be separated into clusters. This introduces bias into models trained using random splitting, where those clusters will be evenly distributed, and allows the models to memorize compounds and subsequently predict their toxicity based on the neighbouring compounds (Riley 2019), resulting in overly optimistic predictive performance (Guo et al. 2024). The chemical space of known molecules is complex, indicating that the chemical space of unknown compounds is even more complex.

The finding that a comparatively basic pipeline achieves discriminative performance above chance demonstrates that structure-based approaches have potential to reliably harness toxicity signals. Toxicity prediction from chemical structures is an active research area where substantial theoretical research is still conducted (see Seal et al. 2025, Monem et al. 2025, Huang and D. D. Wang 2026). However, building a de novo architecture better at detecting the chemical signals and training a model with increased complexity is beyond

the scope of this work.

4.4 Outlook

Several steps are obvious following the approach of this work. First, the random split does not transfer to truly novel compounds, therefore more realistic ways of data splits need to be investigated for this problem, such as scaffold splits (which come with their own limitations, see Kretschmer et al. 2025) or maximum common edge subgraph splits (Raymond et al. 2002), which give more realistic estimates of performance on compounds dissimilar to those seen in training. Second, density-based clustering methods (e.g., DBSCAN) could be better suited to the sparse, non-linear structure of binary fingerprints rather than a variance-based linear method such as PCA. Incorporating the class imbalance into the model directly as a prior rather than through resampling could be evaluated against the here employed SMOTEENN approach. Additionally, other methods of fingerprint generation should be investigated, see Huber and Pollmann 2026 for a comprehensive review on possible representations. Finally, alternative model architectures, such as graph neural networks operating directly on the molecular structure should be considered (see Atz et al. 2021 for an extensive review).

This work presents a primary investigation into the applicability of the PlastChem database for structure-based hazard prediction. It underlines the inherent complexity of chemical machine learning and illustrates that toxicity prediction from molecular structures is a multidimensional problem, restrained by data availability and representation choices as well as model architecture. The scope of this work is too limited to sufficiently solve this problem, but rather serves as a stepping stone for further, more extensive work. Rather than an exact substitute for an experimental hazard assessment, the model presented in this work could serve as a primitive filter for triaging novel chemicals of concern.

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List of Tables

1	Results of iterative grid search, with chosen parameters underlined.	9
2	Classification performance on the test set.	12

List of Figures

1	Amount and ratio of hazardous and non-hazardous compounds in the dataset displayed by polymer type for the ten most abundant polymer types present.	5
2	Two-dimensional principal component analysis of dataset represented as molecular fingerprints, explaining 8.56% of total variance.	6
3	Cumulative explained variance ratio by principal components. Red marker indicates number of components selected by grid search (n=160, corresponding to 59.21% explained variance.)	10
4	Effect of SMOTEENN resampling on the class distribution of the training set.	11
5	Nested cross-validation performance (balanced accuracy across five outer folds) of the pipeline trained with and without SMOTEENN resampling. Dashed black line indicates mean balanced accuracy across all folds. Dashed green line indicates balanced accuracy of test set.	12
6	Precision-recall curves for both classes on the independent test set (n=1425, of which 47 are labelled as non-hazardous).	13
7	ROC curve for the final XGBoost model on the independent test set, compared against a dummy baseline.	14
S1	”Workflow to assign hazard scores. PBT refers to persistent, bioaccumulative and toxic chemicals; vPvB refers to very persistent, very bioaccumulative chemicals; POP refers to persistent organic pollutants; PMT refers to persistent, mobile and toxic chemicals; vPvM refers to very persistent, very mobile chemicals; CMR refers to chemicals that are carcinogenic, mutagenic and toxic to reproduction; EDC refers to endocrine disrupting chemicals; STOT refers to chemicals with specific target organ toxicity (after repeated exposure), Aq. Tox. refers to chemicals inducing aquatic toxicity.” (Monclús et al. 2025, Supplements p. 13)	21
S2	Comparison of the absolute factor loadings of the three additionally included physiochemical properties (Molecular weight, TPSA and complexity) with the maximum factor loading of the Morgan fingerprint features for each of the first ten principal components.	22

S3	Balanced accuracy of the inner cross-validation (used for hyperparameter selection within each outer fold), shown for comparison with the outer nested CV scores in Figure 5. Systematically higher and less variable than the outer scores, illustrating the optimistic selection bias nested CV is designed to correct for.	22
S4	Confusion matrix for the trained model on the independent test set (n=1425).	23
S5	Confusion matrix for the classifier trained without resampling on the independent test set (n=1425).	23
S6	Confusion matrix for the dummy classifier on the independent test set (n=1425).	24

S1 Code availability

All code used for data processing, training, model implementation and visualization can be found under the project thesis Github.

S2 Additional Figure

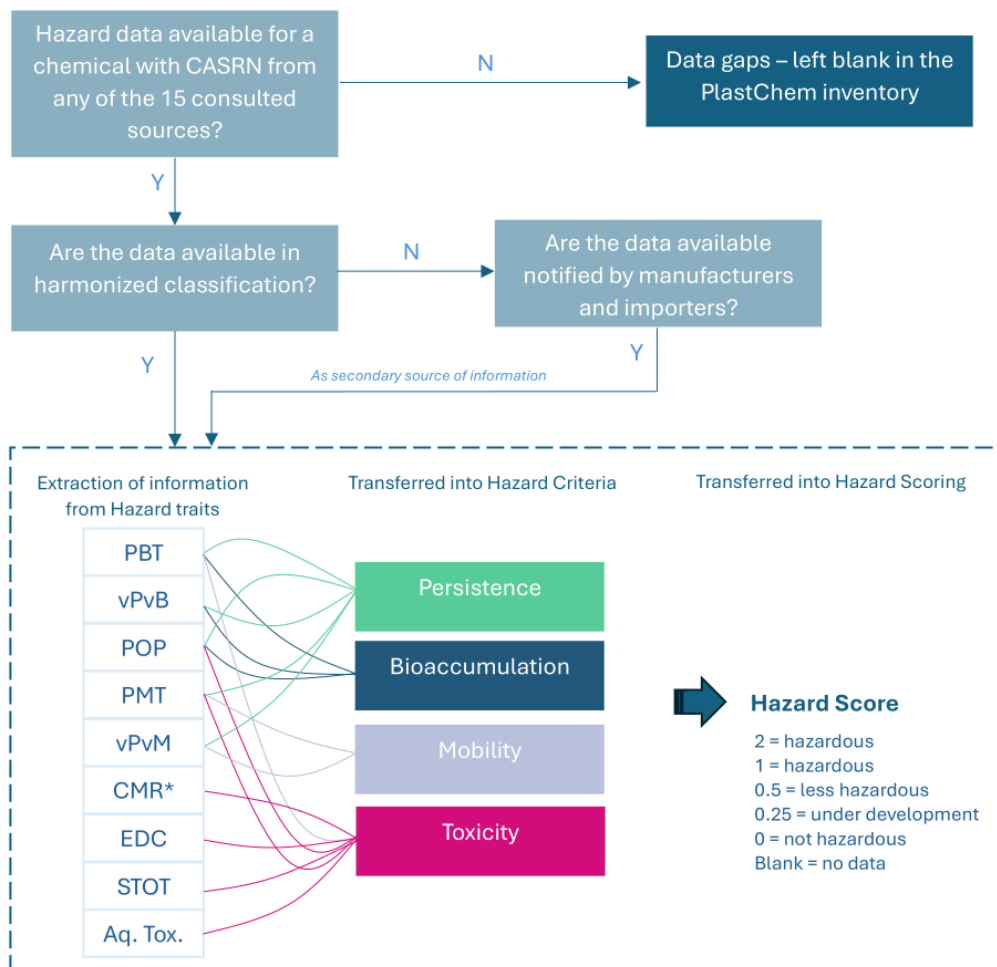


Figure S1: "Workflow to assign hazard scores. PBT refers to persistent, bioaccumulative and toxic chemicals; vPvB refers to very persistent, very bioaccumulative chemicals; POP refers to persistent organic pollutants; PMT refers to persistent, mobile and toxic chemicals; vPvM refers to very persistent, very mobile chemicals; CMR refers to chemicals that are carcinogenic, mutagenic and toxic to reproduction; EDC refers to endocrine disrupting chemicals; STOT refers to chemicals with specific target organ toxicity (after repeated exposure), Aq. Tox. refers to chemicals inducing aquatic toxicity." (Monclús et al. 2025, Supplements p. 13)

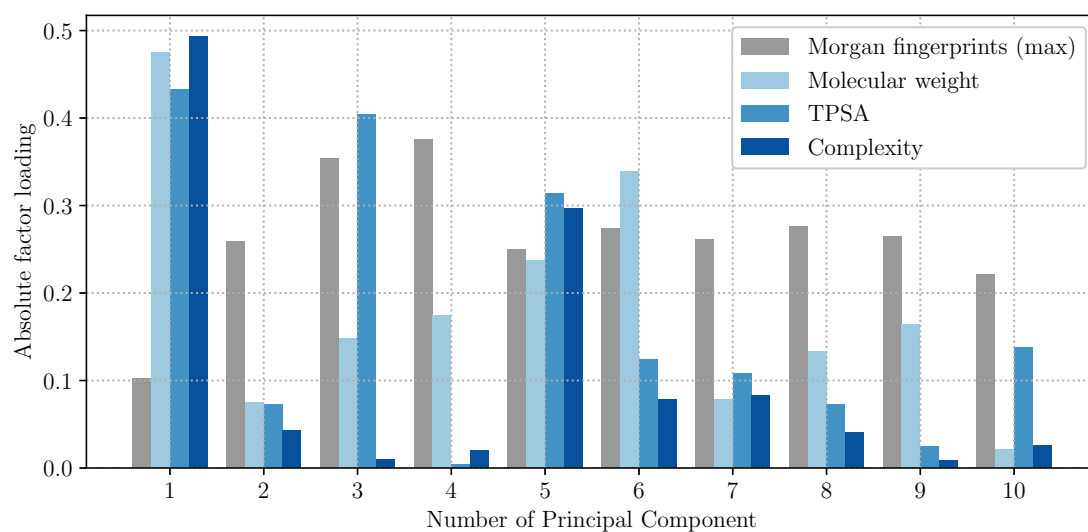


Figure S2: Comparison of the absolute factor loadings of the three additionally included physiochemical properties (Molecular weight, TPSA and complexity) with the maximum factor loading of the Morgan fingerprint features for each of the first ten principal components.

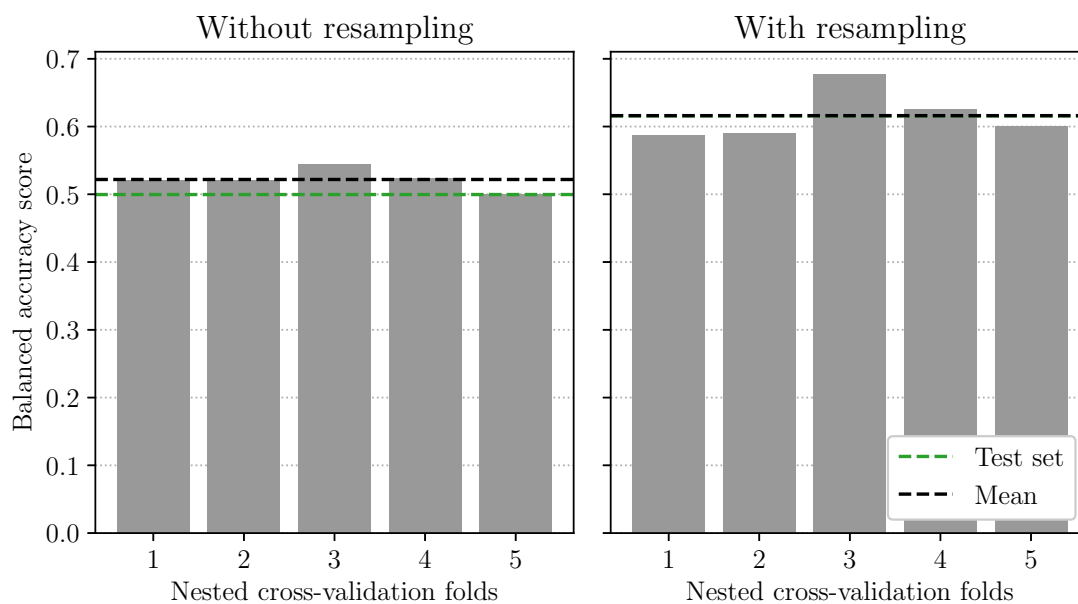


Figure S3: Balanced accuracy of the inner cross-validation (used for hyperparameter selection within each outer fold), shown for comparison with the outer nested CV scores in Figure 5. Systematically higher and less variable than the outer scores, illustrating the optimistic selection bias nested CV is designed to correct for.

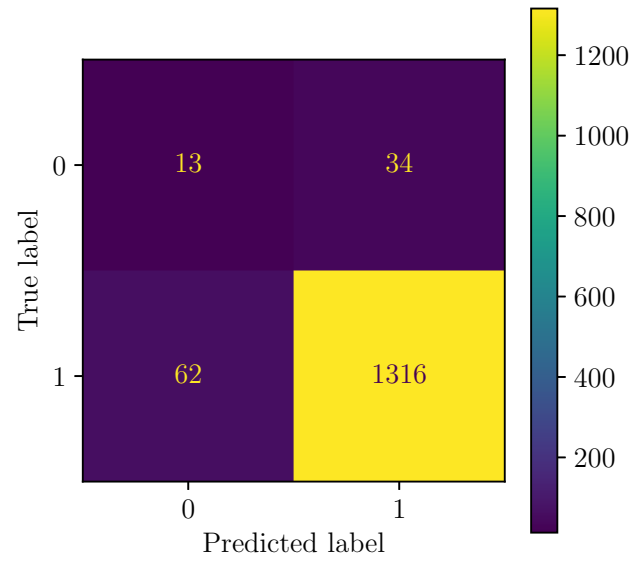


Figure S4: Confusion matrix for the trained model on the independent test set (n=1425).

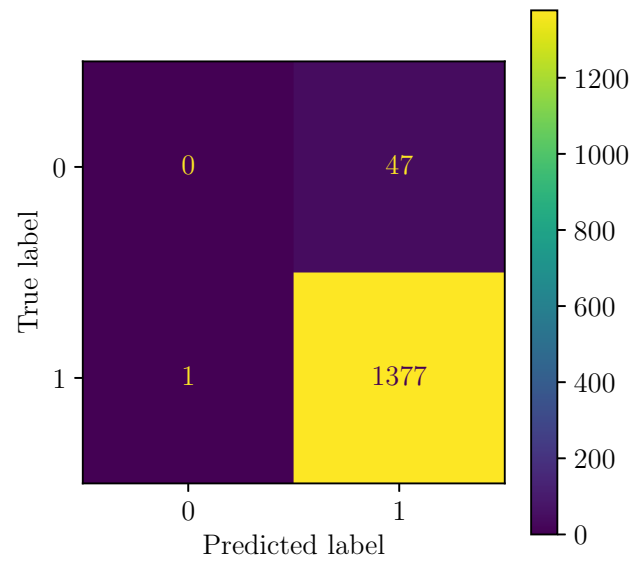


Figure S5: Confusion matrix for the classifier trained without resampling on the independent test set (n=1425).

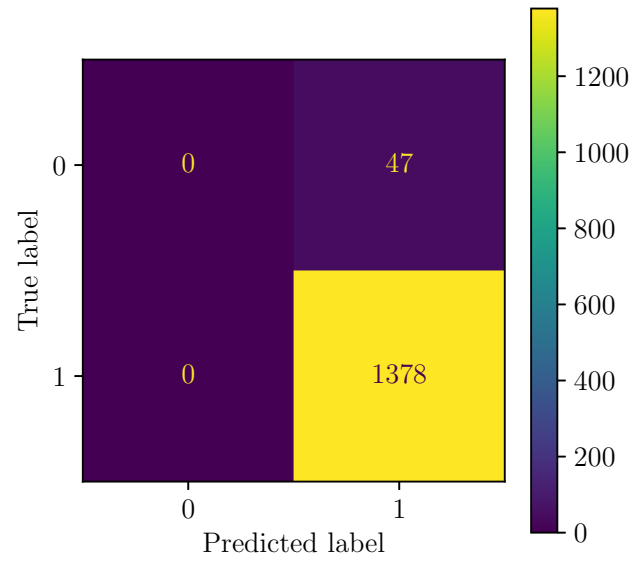


Figure S6: Confusion matrix for the dummy classifier on the independent test set (n=1425).

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Jena, September 23, 2026

Amelie Hübner